Asset Management & Sustainable Finance

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1 Portfolio optimization and risk budgeting

We consider the CAPM model:

$$R_i - r = \beta_i \cdot (R_m - r) + \varepsilon_i$$

where R_i is the return of asset i, R_m is the return of the market portfolio, r is the risk-free asset, β_i is the beta of asset i with respect to the market portfolio and ε_i is the idiosyncratic risk. We assume that $R_m \sim \mathcal{N}(\mu_m, \sigma_m^2)$, $\varepsilon_i \sim \mathcal{N}(0, \tilde{\sigma}_i^2)$, $R_m \perp \varepsilon_i$ and $\varepsilon_i \perp \varepsilon_j$. We denote μ_m as the expected return of the market portfolio, σ_m as the volatility of the market portfolio and $\tilde{\sigma}_i$ as the idiosyncratic volatility.

We consider a universe of 8 assets with the following parameter values:

Asset i	#1	#2	#3	#4	#5	#6	#7	#8
β_i	-1.00	-0.30	0.30	0.60	1.00	1.20	1.50	2.50
$\tilde{\sigma}_i$	15%	16%	10%	11%	8%	13%	12%	14%

and $\sigma_m = 20\%$. The risk-free return is set to 1% and we assume that the expected return of the market portfolio is equal to $\mu_m = 7\%$.

We define the leverage ratio of the portfolio x as the sum of the absolute weights:

$$\mathcal{L}(x) = \sum_{i=1}^{n} |x_i|$$

1.1 Point 1

(a) Compute the vector μ of expected returns

To compute the vector of expected returns μ, we use the Capital Asset Pricing Model (CAPM). According to the CAPM, the expected return of asset i is given by:

$$\mu_i = r + \beta_i(\mu_m - r)$$

where r is the risk-free rate, μ_m is the expected return of the market portfolio and β_i is the sensitivity of asset i to the market portfolio.

In our case we use the following parameters:

$$r = 1\%$$

The market risk premium is therefore:

$$\pi = \mu_m - r = 7\% - 1\% = 6\%$$

Using the CAPM formula, the expected returns μ can be summarized in the following table:

Asset	**Expected Return (μ)**
Asset 1	-5.0%
Asset 2	-0.8%
Asset 3	2.8%
Asset 4	4.6%
Asset 5	7.0%
Asset 6	8.2%
Asset 7	10.0%
Asset 8	16.0%

Table 1: Vector of expected returns estimated using the CAPM

(b) Compute the covariance matrix Σ of stock returns.

Under the CAPM assumptions, the return of asset i can be written as

$$R_i = r + \beta_i(R_m - r) + \varepsilon_i$$

where ε_i represents the idiosyncratic risk. Since the idiosyncratic components are independent across assets and independent from the market factor, the covariance matrix of asset returns can be written as

$$\Sigma = \sigma_m^2 \, \beta\beta^\top + \mathrm{diag}(\tilde{\sigma}_1^2, \ldots, \tilde{\sigma}_8^2)$$

where σ_m is the market volatility, β is the vector of betas and $\tilde{\sigma}_i$ are the idiosyncratic volatilities. Solving with $\sigma_m = 20\%$, $\beta = (-1.00, -0.30, 0.30, 0.60, 1.00, 1.20, 1.50, 2.50)$ and $\tilde{\sigma} = (15\%, 16\%, 10\%, 11\%, 8\%, 13\%, 12\%, 14\%)$, we obtain the following covariance matrix:

$$\Sigma = \begin{pmatrix} 6.25\% & 1.20\% & -1.20\% & -2.40\% & -4.00\% & -4.80\% & -6.00\% & -10.00\% \\ 1.20\% & 2.92\% & -0.36\% & -0.72\% & -1.20\% & -1.44\% & -1.80\% & -3.00\% \\ -1.20\% & -0.36\% & 1.36\% & 0.72\% & 1.20\% & 1.44\% & 1.80\% & 3.00\% \\ -2.40\% & -0.72\% & 0.72\% & 2.65\% & 2.40\% & 2.88\% & 3.60\% & 6.00\% \\ -4.00\% & -1.20\% & 1.20\% & 2.40\% & 4.64\% & 4.80\% & 6.00\% & 10.00\% \\ -4.80\% & -1.44\% & 1.44\% & 2.88\% & 4.80\% & 7.45\% & 7.20\% & 12.00\% \\ -6.00\% & -1.80\% & 1.80\% & 3.60\% & 6.00\% & 7.20\% & 10.44\% & 15.00\% \\ -10.00\% & -3.00\% & 3.00\% & 6.00\% & 10.00\% & 12.00\% & 15.00\% & 26.96\% \end{pmatrix}$$

(c) Deduce the vector σ of volatilities and the correlation matrix ρ of stock returns.

The vector of volatilities σ is obtained from the covariance matrix Σ by taking the square root of its diagonal elements:

$$\sigma = \sqrt{diag(\Sigma)} \quad \Rightarrow \quad \sigma_i = \sqrt{\Sigma_{ii}}$$

This gives the following volatilities:

Asset	**Volatility (σ)**
Asset 1	25.00%
Asset 2	17.09%
Asset 3	11.66%
Asset 4	16.28%
Asset 5	21.54%
Asset 6	27.29%
Asset 7	32.31%
Asset 8	51.92%

Table 2: Vector of asset volatilities

The correlation matrix ρ is computed as:

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}$$

which yields:

$$\rho = \begin{pmatrix} 100.00\% & 28.09\% & -41.16\% & -58.97\% & -74.28\% & -70.34\% & -74.28\% & -77.04\% \\ 28.09\% & 100.00\% & -18.07\% & -25.88\% & -32.60\% & -30.87\% & -32.60\% & -33.81\% \\ -41.16\% & -18.07\% & 100.00\% & 37.93\% & 47.77\% & 45.24\% & 47.77\% & 49.54\% \\ -58.97\% & -25.88\% & 37.93\% & 100.00\% & 68.44\% & 64.82\% & 68.44\% & 70.99\% \\ -74.28\% & -32.60\% & 47.77\% & 68.44\% & 100.00\% & 81.64\% & 86.21\% & 89.41\% \\ -70.34\% & -30.87\% & 45.24\% & 64.82\% & 81.64\% & 100.00\% & 81.64\% & 84.67\% \\ -74.28\% & -32.60\% & 47.77\% & 68.44\% & 86.21\% & 81.64\% & 100.00\% & 89.41\% \\ -77.04\% & -33.81\% & 49.54\% & 70.99\% & 89.41\% & 84.67\% & 89.41\% & 100.00\% \end{pmatrix}$$

(d) Compute the Sharpe ratio of each asset.

The Sharpe ratio of asset i is defined as

$$SR_i = \frac{\mu_i - r}{\sigma_i}$$

Using the expected returns μ, the risk-free rate $r = 1\%$, and the volatilities σ, we obtain the following Sharpe ratios:

Asset	Sharpe Ratio
1	-0.2400
2	-0.1053
3	0.1543
4	0.2211
5	0.2785
6	0.2638
7	0.2785
8	0.2889

(e) How do you interpret the first two assets #1 and #2? Compute the correlation between asset #1 and the equally-weighted portfolio based on the last six assets. Comment on these results.

Asset #1 and Asset #2 have negative betas ($\beta_1 = -1.0$ and $\beta_2 = -0.3$). Therefore, their returns tend to move in the opposite direction of the market portfolio.

To illustrate this effect, we compute the correlation between Asset #1 and an equally-weighted portfolio composed of assets #3 to #8, whose assets have positive β_i. The equally-weighted portfolio (ewp) is defined with weights

$$w_{ewp} = \left(0, 0, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}\right)$$

The resulting correlation is

$$\rho_{1,ewp} = -78.47\%.$$

This strong negative correlation shows that Asset #1 moves in the opposite direction of the diversified portfolio composed of the other assets, providing significant diversification benefits.

1.2 Point 2

We consider long/short MVO portfolios such that $\sum_{i=1}^{n} x_i = 1$.

(a) Give the QP formulation of the mean-variance optimization problem.

$$x^\star = \arg\min \left(\frac{1}{2} x^\top \Sigma x - \gamma x^\top (\mu - r\mathbf{1}_8) \right)$$

$$\text{s.t.} \begin{cases} \sum_{i=1}^{8} x_i = 1, \\ -10 \leq x_i \leq 10, \quad i = 1, \ldots, 8. \end{cases}$$

A standard quadratic programming problem can be written as

$$x^\star = \arg\min \left(\frac{1}{2} x^\top Q x - x^\top R \right)$$

$$\text{s.t} \quad \begin{cases} Ax = B \\ Cx \leq D \\ x^- \leq x \leq x^+ \end{cases}$$

subject to linear equality and inequality constraints. In our case, the quadratic term is given by the covariance matrix of returns and the linear term is given by the excess expected return vector:

$$Q = \Sigma, \qquad R = \gamma(\mu - r\mathbf{1}_8),$$

while the constraints are given by:

$$A = \mathbf{1}_8^\top, \qquad B = 1$$

$$x^- = (-10, \ldots, -10)^\top, \qquad x^+ = (10, \ldots, 10)^\top$$

where $\mathbf{1}_n \in \mathbb{R}^n$ is the vector of ones of length n. In this point there are no additional inequality constraints $Cx \leq D$.

(b) Using the γ-problem, find the optimal solution $x^\star(\gamma)$ when the coefficient γ is equal to 0, 0.10, 0.25, 0.50, 0.75, and 1.00. For each optimized portfolio, compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$, its Sharpe ratio $SR(x^\star(\gamma) \mid r)$ and the corresponding leverage ratio $\mathcal{L}(x^\star(\gamma))$.

For each value of γ, we solve the quadratic program and obtain the optimal weights $x^\star(\gamma)$. We then compute the portfolio expected return $\mu_p = \mu(x^\star(\gamma)) = \mu^\top x^\star(\gamma)$, volatility $\sigma_p = \sigma(x^\star(\gamma)) = \sqrt{x^\star(\gamma)^\top \Sigma x^\star(\gamma)}$, Sharpe ratio $SR_p = SR(x^\star(\gamma) \mid r) = (\mu_p - r)/\sigma_p$ and leverage ratio $\mathcal{L}_p = \mathcal{L}(x^\star(\gamma)) = \sum_{i=1}^{8} |x_i^\star(\gamma)|$. The results are summarized in Table 3.

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	μ_p	σ_p	SR_p	$\mathcal{L}_p$
0.00	21.90%	14.29%	25.70%	16.74%	20.32%	5.55%	2.73%	-7.25%	1.27%	5.58%	0.0488	1.1449
0.10	19.18%	12.83%	24.00%	16.18%	21.38%	6.35%	4.38%	-4.30%	2.12%	6.30%	0.1781	1.0861
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%	3.39%	9.18%	0.2610	1.0000
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%	5.52%	15.60%	0.2895	1.0000
0.75	1.52%	3.34%	12.94%	12.51%	28.23%	11.56%	15.09%	14.83%	7.64%	22.56%	0.2944	1.0000
1.00	-5.28%	-0.32%	8.68%	11.10%	30.86%	13.56%	19.21%	22.18%	9.76%	29.67%	0.2954	1.1119

Table 3: Optimal portfolio weights $x^\star(\gamma)$ and corresponding portfolio statistics for selected values of γ.

(c) Draw the efficient frontier by considering granular values of γ.

To draw the efficient frontier, we solve the γ-problem on a fine grid of risk-tolerance parameters γ. In particular, we consider

$$\gamma \in \{-0.50, -0.45, -0.40, \ldots, -0.10, -0.05, 0, 0.05, 0.10, \ldots, 0.95, 1, 2, \ldots, 10\}.$$

. For each γ, we obtain the optimal portfolio $x^\star(\gamma)$ and compute its expected return

$$\mu_p(\gamma) = \mu^\top x^\star(\gamma)$$

and its volatility

$$\sigma_p(\gamma) = \sqrt{x^\star(\gamma)^\top \Sigma x^\star(\gamma)}.$$

Plotting the pairs $\big(\sigma_p(\gamma), \mu_p(\gamma)\big)$ gives the efficient frontier.

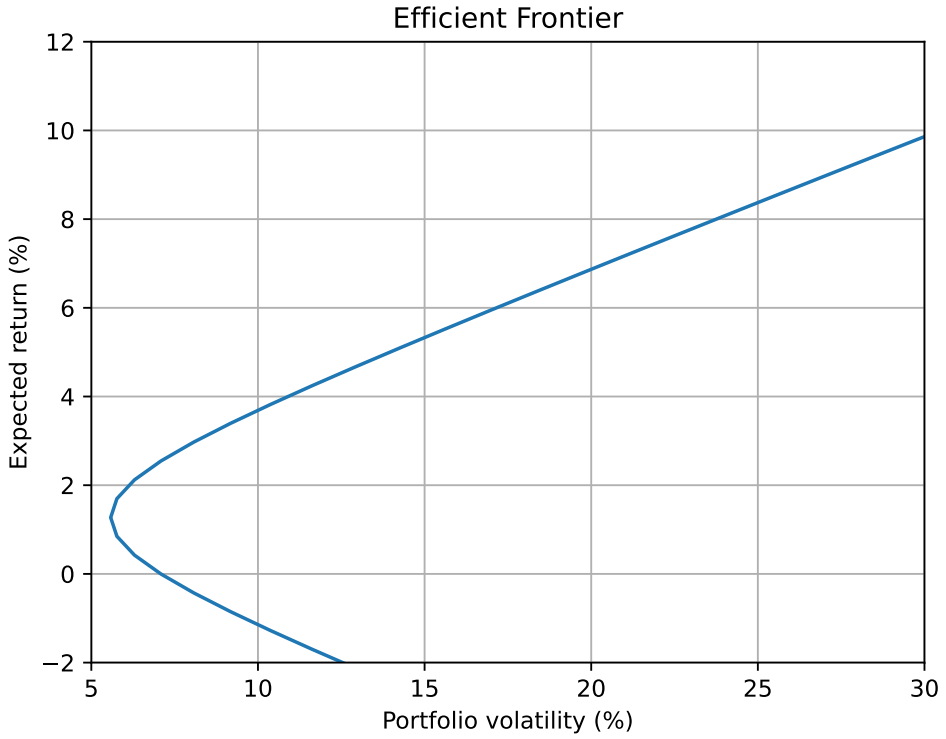


Figure 1: Efficient frontier obtained by solving the γ-problem for granular values of γ.

The left-most point corresponds to the minimum-variance portfolio, while moving along the upper branch increases expected return together with risk. The lower branch represents inefficient portfolios, since for the same level of risk a higher expected return can be achieved on the upper branch.

(d) Draw the relationship between the risk tolerance γ and the leverage ratio $\mathcal{L}(x^\star(\gamma))$ for $\gamma \in [-0.5, 2.0]$. Comment on these results.

To analyze the relationship between risk tolerance and leverage, we solve the γ-problem for values of γ in the interval $[-0.5, 2.0]$. For each optimal portfolio $x^\star(\gamma)$, we compute the leverage ratio as

$$\mathcal{L}(x^\star(\gamma)) = \sum_{i=1}^{8} |x_i^\star(\gamma)|.$$

The resulting relationship between γ and $\mathcal{L}(x^\star(\gamma))$ is shown in Figure 2.

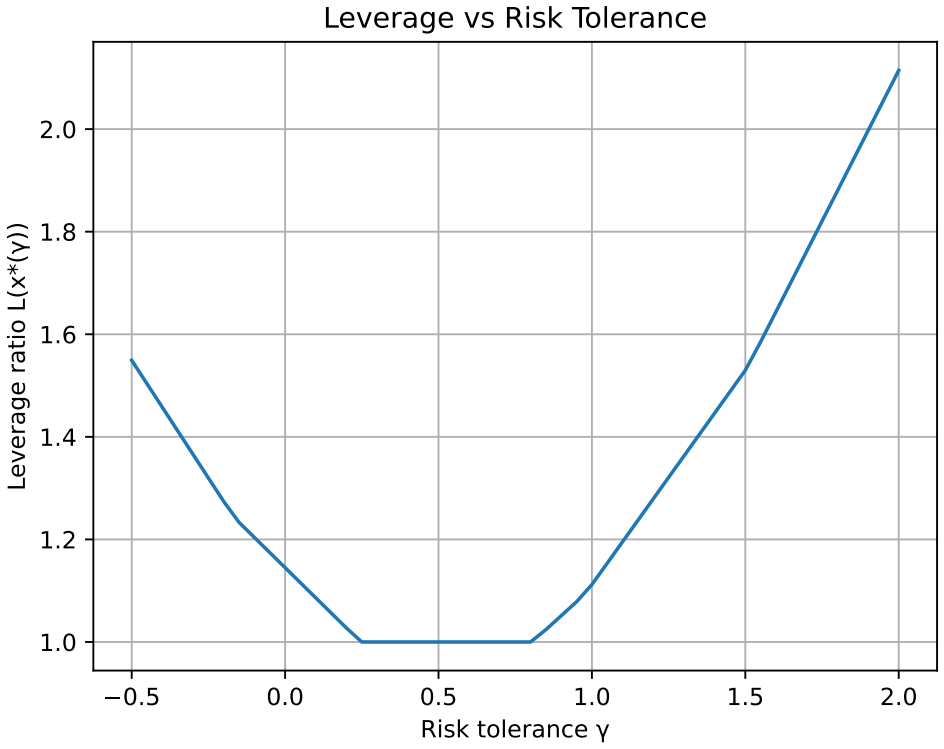


Figure 2: Leverage ratio $\mathcal{L}(x^\star(\gamma))$ as a function of the risk tolerance parameter γ.

Around $\gamma \approx 0$, the objective is dominated by variance minimization. The optimal portfolio is close to the minimum-variance portfolio, with limited short positions and nearly full investment, implying $\mathcal{L} \approx 1$.

For intermediate values of γ (approximately $\gamma \in [0.2, 0.8]$), the leverage ratio remains equal to one, while for $\gamma > 0.8$, the expected return term becomes progressively more relevant. The optimizer tilts the portfolio towards high-return assets, which under the constraint $\sum_{i=1}^{8} x_i = 1$ leads to larger long and short positions. As a result, the leverage ratio $\mathcal{L}$ increases with γ.

For $\gamma < 0$, the expected return term is effectively penalized:

$$x^\star = \arg\min \left(\frac{1}{2} x^\top \Sigma x + \gamma\, x^\top (\mu - r\mathbf{1}) \right).$$

The optimizer tends to short high-return assets and go long low-return assets. This again requires offsetting positions, leading to $\mathcal{L} > 1$.

(e) Using a bisection algorithm, calculate the MVO portfolio if we target an ex-ante volatility of 10% and 15%. Give the corresponding value of γ of the QP problem. For each optimized portfolio, compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$ and its Sharpe ratio $SR(x^\star(\gamma) \mid r)$.

We use a bisection algorithm on γ to find $\gamma^\star$ such that the optimal solution $x^\star(\gamma^\star)$ satisfies the target volatility $\sigma(x^\star(\gamma^\star))$. The algorithm iteratively halves the interval of possible values of γ and selects the subinterval whose portfolio volatility is closer to the target until convergence.

Target σ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
10%	14.16%	10.13%	20.86%	15.14%	23.32%	7.83%	7.43%	1.13%
15%	8.91%	7.31%	17.57%	14.05%	25.36%	9.38%	10.61%	6.82%

Table 4: Optimal weights $x^\star(\gamma^\star)$ for each volatility target.

Target σ	$\gamma^\star$	$\mu(x^\star(\gamma^\star))$	$\sigma(x^\star(\gamma^\star))$	$SR(x^\star(\gamma^\star) \mid r)$
10%	0.2848	3.69%	10.00%	0.2690
15%	0.4778	5.33%	15.00%	0.2886

Table 5: Bisection results for volatility targets.

As expected, targeting a higher volatility (15%) requires a higher risk tolerance parameter ($\gamma^\star$ increases) and leads to a higher expected return and Sharpe ratio compared to the 10% target.

(f) Using the analytical formula, find the tangency portfolio. Compute its expected return $\mu(x^\star)$, its volatility $\sigma(x^\star)$ and its Sharpe ratio $SR(x^\star \mid r)$. What is the value of the implied risk tolerance associated to the tangency portfolio?

The tangency portfolio is obtained analytically as

$$z = \Sigma^{-1}(\mu - r\mathbf{1}_8), \qquad x_{\tan} = x^\star = \frac{z}{\mathbf{1}_8^\top z},$$

which satisfies the budget constraint $\sum_{i=1}^8 x_i = 1$. The implied risk tolerance γ_{impl} associated to the tangency portfolio is

$$\phi_0 = \mathbf{1}_8^\top \Sigma^{-1}(\mu - r\mathbf{1}_8), \qquad \gamma_{\text{impl}} = \phi_0^{-1}.$$

Asset	$x_{\tan}$
1	-9.21%
2	-2.43%
3	6.22%
4	10.28%
5	32.39%
6	14.72%
7	21.59%
8	26.44%

Table 6: Tangency portfolio weights (analytical solution).

γ_{impl}	$\mu(x_{\tan})$	$\sigma(x_{\tan})$	$SR(x_{\tan} \mid r)$
1.1447	10.99%	33.82%	0.2954

Table 7: Tangency portfolio performance and implied risk tolerance.

The tangency portfolio maximizes the Sharpe ratio among fully-invested portfolios under the model assumptions.

(g) Using the efficient frontier with a fine grid of γ, find the tangency portfolio using the brute force algorithm. Compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$ and its Sharpe ratio $SR(x^\star(\gamma) \mid r)$.

We compute the efficient frontier on a fine grid of γ values and select the portfolio that maximizes the Sharpe ratio:

$$\gamma^\star = \arg\max_{\gamma} \; SR(x^\star(\gamma) \mid r)\,.$$

The resulting tangency portfolio (brute force) is reported below.

Asset	$x^\star(\gamma^\star)$
x_1	-9.21%
x_2	-2.43%
x_3	6.22%
x_4	10.28%
x_5	32.39%
x_6	14.72%
x_7	21.59%
x_8	26.44%

Table 8: Brute-force tangency portfolio weights.

$\gamma^\star$	$\mu(x^\star(\gamma^\star))$	$\sigma(x^\star(\gamma^\star))$	$SR(x^\star(\gamma^\star) \mid r)$
1.1447	10.99%	33.82%	0.2954

Table 9: Tangency portfolio found by brute force on a fine grid of γ.

This brute-force tangency portfolio matches the analytical tangency portfolio, confirming that maximizing the Sharpe ratio along the efficient frontier recovers the tangency solution.

(h) Now consider the extended quadratic programming problem by including the risk-free asset in the investment universe. Formulate the extended QP problem, and solve it using a fine grid of $\gamma \in [0.5, 1.5]$. What do we observe when calculating the Sharpe ratio of the different optimized portfolios? How can this be explained? Which criterion can be used to implement the brute-force algorithm on the extended QP problem to find the tangency portfolio?

We extend the investment universe by including the risk-free asset. The extended weight vector is

$$x_{\text{ext}} = \big(x_1, \cdots, x_8, x_{rf}\big)^\top \in \mathbb{R}^9,$$

where x_{rf} is the weight invested in the risk-free asset.

The extended QP problem corresponding to the general QP formulation introduced in Point 2(a) is:

$$x^\star_{\text{ext}}(\gamma) = \arg\min_{x_{\text{ext}}} \left(\frac{1}{2} x_{\text{ext}}^\top Q x_{\text{ext}} - x_{\text{ext}}^\top R \right)$$

$$\text{s.t.} \begin{cases} A x_{\text{ext}} = B, \\ -10\,\mathbf{1}_9 \leq x_{\text{ext}} \leq 10\,\mathbf{1}_9. \end{cases}$$

where the quadratic terms, the linear term and the constraints are identified as:

$$Q = \Sigma_{\text{ext}} = \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix}. \qquad R = \gamma(\mu_{\text{ext}} - r\mathbf{1}_9), \qquad \mu_{\text{ext}} = \begin{pmatrix} \mu \\ r \end{pmatrix},$$

$$A = \mathbf{1}_9^\top, \qquad B = 1.$$

Solving the problem on a fine grid $\gamma \in [0.5, 1.5]$, we observe that the Sharpe ratio remains numerically constant (around 0.2954) across the optimized portfolios. In particular, computing the average Sharpe ratio over all portfolios in this range yields a mean value of 0.2954 with zero variance, confirming that the Sharpe ratio does not vary across the considered values of γ. This result can also be verified graphically: as shown in Figure 3, the Sharpe ratio plotted as a function of γ is essentially flat.

This occurs because once the risk-free asset is included, all mean-variance optimal portfolios lie on the Capital Market Line. Therefore they are combinations of the risk-free asset and the tangency portfolio, which implies that they all share the same maximal Sharpe ratio.

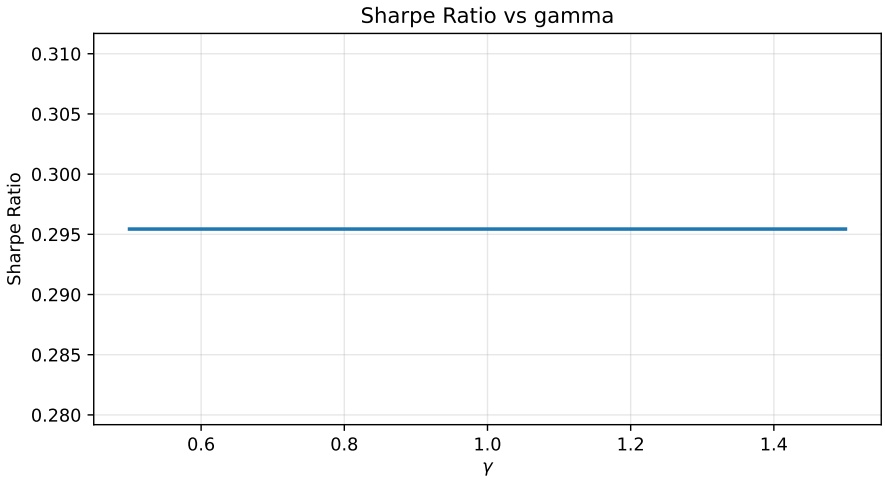


Figure 3: Sharpe ratio as a function of the risk tolerance parameter γ.

Tables 10 and 11 present a sample of computed portfolios, reporting respectively the corresponding asset weights and the portfolio statistics.

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_{rf}
0.5000	-4.02%	-1.06%	2.72%	4.49%	14.15%	6.43%	9.43%	11.55%	56.32%
0.5001	-4.02%	-1.06%	2.72%	4.49%	14.15%	6.43%	9.43%	11.55%	56.31%
0.5002	-4.03%	-1.06%	2.72%	4.49%	14.15%	6.43%	9.44%	11.55%	56.30%
1.4998	-12.07%	-3.18%	8.15%	13.47%	42.44%	19.29%	28.29%	34.64%	-31.02%
1.4999	-12.07%	-3.18%	8.15%	13.47%	42.44%	19.29%	28.29%	34.65%	-31.03%
1.5000	-12.07%	-3.18%	8.15%	13.47%	42.44%	19.29%	28.30%	34.65%	-31.04%

Table 10: Portfolio weights corresponding to the selected γ values.

γ	μ_p	σ_p	SR
0.5000	5.36%	14.77%	0.2954
0.5001	5.36%	14.77%	0.2954
0.5002	5.37%	14.78%	0.2954
1.4998	14.09%	44.31%	0.2954
1.4999	14.09%	44.31%	0.2954
1.5000	14.09%	44.32%	0.2954

Table 11: Subset of portfolios obtained from the extended QP problem.

Since the Sharpe ratio is constant, it cannot be used to identify the tangency portfolio in the extended problem. A practical brute-force criterion is to select the portfolio with risk-free weight closest to zero:

$$\gamma^\star = \arg \min_{\gamma \in [0.5, 1.5]} \left| x^\star_{rf}(\gamma) \right|.$$

The resulting tangency portfolio $x^\star$ is reported in Table 12 and Table 13

Asset	$x^\star$
x_1	-9.21%
x_2	-2.43%
x_3	6.22%
x_4	10.28%
x_5	32.39%
x_6	14.72%
x_7	21.59%
x_8	26.44%
x_{rf}	0.00%

Table 12: Weights of the assets in the tangency portfolio

$\gamma^\star$	$\mu(x^\star)$	$\sigma(x^\star)$	$SR(x^\star \mid r)$
1.1447	10.99%	33.82%	0.2954

Table 13: Tangency portfolio statistics

(i) Compare the three solutions (f), (g) and (h) in terms of risk tolerance γ, weights $x^\star(\gamma)$, expected return $\mu(x^\star(\gamma))$, volatility $\sigma(x^\star(\gamma))$ and Sharpe ratio $SR(x^\star(\gamma) \mid r)$.

Method	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_{rf}
(f) Analytical	−9.21%	−2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%	0.00%
(g) Brute force	−9.21%	−2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%	0.00%
(h) Extended QP	−9.21%	−2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%	0.00%

Table 14: Risky-asset weights obtained with the three methods.

Method	γ	$\mu(x^\star(\gamma))$	$\sigma(x^\star(\gamma))$	$SR(x^\star(\gamma) \mid r)$
(f) Analytical	1.1447	10.99%	33.82%	0.2954
(g) Brute force	1.1447	10.99%	33.82%	0.2954
(h) Extended QP	1.1447	10.99%	33.82%	0.2954

Table 15: Comparison of the tangency portfolio obtained with the three methods.

The three methods produce essentially the same tangency portfolio. The values of γ, expected return, volatility, Sharpe ratio and weight are identical. This confirms that the analytical solution, the brute-force search on the efficient frontier and and the extended QP formulation solve the same optimization problem and therefore identify the same tangency portfolio.

1.3 Point 3

We consider long/short MVO portfolios such that $\sum_{i=1}^{n} x_i = 1$ and $\mathcal{L}(x) = 1$.

(a) We consider the decomposition $x = x^{\text{long}} - x^{\text{short}}$ with $x^{\text{long}} \geq 0$ and $x^{\text{short}} \geq 0$. Write the mean-variance optimization problem in terms of x, x^{long} and x^{short}.

Introduce two nonnegative vectors $x^{\text{long}}, x^{\text{short}} \in \mathbb{R}^8_+$ such that

$$x = x^{\text{long}} - x^{\text{short}}.$$

Then the leverage constraint becomes

$$\mathcal{L}(x) = \sum_{i=1}^{8} |x_i| = \mathbf{1}_8^\top x^{\text{long}} + \mathbf{1}_8^\top x^{\text{short}} = 1,$$

and the budget constraint can be written as

$$\sum_{i=1}^{8} x_i = \mathbf{1}_8^\top (x^{\text{long}} - x^{\text{short}}) = 1.$$

The mean-variance optimization problem is therefore

$$\min_{x^{\text{long}},\, x^{\text{short}}} \left(\frac{1}{2} (x^{\text{long}} - x^{\text{short}})^\top \Sigma (x^{\text{long}} - x^{\text{short}}) - \gamma \, (x^{\text{long}} - x^{\text{short}})^\top (\mu - r\mathbf{1}_8) \right)$$

$$\text{s.t.} \quad \begin{cases} \mathbf{1}_8^\top (x^{\text{long}} - x^{\text{short}}) = 1, \\ \mathbf{1}_8^\top x^{\text{long}} + \mathbf{1}_8^\top x^{\text{short}} = 1, \\ x^{\text{long}} \geq 0, \ x^{\text{short}} \geq 0, \\ -10 \leq x_i \leq 10. \end{cases}$$

(b) Give the augmented QP formulation of the previous mean-variance optimization problem.

Introduce the augmented variable

$$z = \begin{pmatrix} x^{\text{long}} \\ x^{\text{short}} \end{pmatrix} \in \mathbb{R}^{16},$$

where the portfolio weights are defined by

$$x = x^{\text{long}} - x^{\text{short}}, \qquad x^{\text{long}} \geq 0, \quad x^{\text{short}} \geq 0.$$

Identification of Q and R To match the general quadratic programming formulation

$$z^\star = \arg\min_z \left(\frac{1}{2} z^\top Q z - z^\top R \right),$$

we expand the quadratic term

$$(x^{\text{long}} - x^{\text{short}})^\top \Sigma (x^{\text{long}} - x^{\text{short}}).$$

Using the identity

$$(a-b)^\top \Sigma (a-b) = a^\top \Sigma a - 2a^\top \Sigma b + b^\top \Sigma b,$$

we obtain

$$x^{\text{long}\top}\Sigma x^{\text{long}} - 2x^{\text{long}\top}\Sigma x^{\text{short}} + x^{\text{short}\top}\Sigma x^{\text{short}}.$$

This expression can be written in quadratic form as

$$z^\top \begin{pmatrix} \Sigma & -\Sigma \\ -\Sigma & \Sigma \end{pmatrix} z,$$

which yields

$$Q = \begin{pmatrix} \Sigma & -\Sigma \\ -\Sigma & \Sigma \end{pmatrix}.$$

The linear term is

$$-\gamma(x^{\text{long}} - x^{\text{short}})^\top(\mu - r\mathbf{1}_8).$$

Separating the terms gives

$$-\gamma x^{\text{long}\top}(\mu - r\mathbf{1}) + \gamma x^{\text{short}\top}(\mu - r\mathbf{1}_8).$$

Writing this in the form $-z^\top R$ gives

$$R = \begin{pmatrix} \gamma(\mu - r\mathbf{1}_8) \\ -\gamma(\mu - r\mathbf{1}_8) \end{pmatrix}.$$

Final augmented QP problem The optimization problem can be written in the standard quadratic programming form

$$z^\star(\gamma) = \arg\min_z \left(\frac{1}{2} z^\top Q z - z^\top R \right)$$

$$\text{s.t.} \quad \begin{cases} Az = B \\ Cz \leq D \\ z^- \leq z \leq z^+ \end{cases}$$

where

$$Q = \begin{pmatrix} \Sigma & -\Sigma \\ -\Sigma & \Sigma \end{pmatrix}, \qquad R = \begin{pmatrix} \gamma(\mu - r\mathbf{1}_8) \\ -\gamma(\mu - r\mathbf{1}_8) \end{pmatrix}.$$

The equality constraints correspond to the budget and leverage constraints:

$$A = \begin{pmatrix} \mathbf{1}_8^\top & -\mathbf{1}_8^\top \\ \mathbf{1}_8^\top & \mathbf{1}_8^\top \end{pmatrix}, \qquad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The inequality constraints enforce both non-negativity and box constraints:

$$C = \begin{pmatrix} -I_8 & 0 \\ 0 & -I_8 \\ I_8 & -I_8 \\ -I_8 & I_8 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 \\ 0 \\ 10\mathbf{1}_8 \\ 10\mathbf{1}_8 \end{pmatrix}.$$

(c) Using the γ-problem, find the optimal solution $x^\star(\gamma)$ when the coefficient γ is equal to 0, 0.10, 0.25, 0.50, 0.75, and 1.00. For each optimized portfolio, compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$, its Sharpe ratio $SR(x^\star(\gamma) \mid r)$ and the corresponding leverage ratio $\mathcal{L}(x^\star(\gamma))$.

For each value of $\gamma \in \{0, 0.10, 0.25, 0.50, 0.75, 1.00\}$ we solve the γ-problem using the augmented QP formulation under the leverage constraint $\mathcal{L}(x) = 1$.

For each optimal portfolio $x^\star(\gamma)$ we compute the corresponding portfolio statistics:

$$\mu_p = \mu(x^\star(\gamma)) = \mu^\top x^\star(\gamma), \qquad \sigma_p = \sigma(x^\star(\gamma)) = \sqrt{x^\star(\gamma)^\top \Sigma x^\star(\gamma)},$$

$$SR_p = SR(x^\star(\gamma) \mid r) = \frac{\mu_p - r}{\sigma_p}, \qquad \mathcal{L}_p = \mathcal{L}(x^\star(\gamma)) = \sum_{i=1}^{8} |x_i^\star(\gamma)|.$$

The resulting optimal portfolios and their statistics are reported in Tables 16 and 17.

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
0.00	25.03%	15.60%	25.92%	15.63%	14.93%	2.89%	0.00%	0.00%
0.10	20.91%	13.57%	24.20%	15.64%	18.57%	4.95%	2.15%	0.00%
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%
0.75	1.52%	3.34%	12.94%	12.51%	28.23%	11.56%	15.09%	14.83%
1.00	0.00%	0.00%	3.86%	8.37%	28.87%	13.41%	20.08%	25.41%

Table 16: Optimal portfolios $x^\star(\gamma)$ under the leverage constraint $\mathcal{L}(x) = 1$.

γ	μ_p	σ_p	SR_p	$\mathcal{L}_p$
0.00	1.35%	5.74%	0.0611	1.00
0.10	2.16%	6.41%	0.1815	1.00
0.25	3.39%	9.18%	0.2610	1.00
0.50	5.52%	15.60%	0.2895	1.00
0.75	7.64%	22.56%	0.2944	1.00
1.00	9.69%	29.43%	0.2951	1.00

Table 17: Portfolio statistics corresponding to the optimal portfolios $x^\star(\gamma)$.

As expected, the leverage ratio remains equal to 1 for all portfolios because of the imposed constraint $\mathcal{L}(x) = 1$.

(d) Compare the two solutions 2(b) and 3(c). Comment on these results.

We compare the optimal portfolios obtained in point 2(b) and point 3(c). Table 18 and Table 19 report the portfolio weights, while Table 20 reports the corresponding portfolio statistics.

The two solutions are very similar in terms of expected return, volatility and Sharpe ratio. The main difference concerns the leverage ratio: in point 2(b), leverage is not constrained, so the optimizer can take larger long/short positions, leading to $\mathcal{L}(x) > 1$ for $\gamma = 0.00$, 0.10 and 1.00. In point 3(c), the constraint $\mathcal{L}(x) = 1$ is imposed, so leverage is fixed at 1 for all portfolios.

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
0.00	21.90%	14.29%	25.70%	16.74%	20.32%	5.55%	2.73%	-7.25%
0.10	19.18%	12.83%	24.00%	16.18%	21.38%	6.35%	4.38%	-4.30%
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%
0.75	1.52%	3.34%	12.94%	12.51%	28.23%	11.56%	15.09%	14.83%
1.00	-5.28%	-0.32%	8.68%	11.10%	30.86%	13.56%	19.21%	22.18%

Table 18: Optimal portfolio weights from point 2(b).

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
0.00	25.03%	15.60%	25.92%	15.63%	14.93%	2.89%	0.00%	0.00%
0.10	20.91%	13.57%	24.20%	15.64%	18.57%	4.95%	2.15%	0.00%
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%
0.75	1.52%	3.34%	12.94%	12.51%	28.23%	11.56%	15.09%	14.83%
1.00	0.00%	0.00%	3.86%	8.37%	28.87%	13.41%	20.08%	25.41%

Table 19: Optimal portfolio weights from point 3(c).

γ	μ_{2b}	σ_{2b}	SR_{2b}	L_{2b}	μ_{3c}	σ_{3c}	SR_{3c}	L_{3c}
0.00	1.27%	5.58%	0.0488	1.1449	1.35%	5.74%	0.0611	1.0000
0.10	2.12%	6.30%	0.1781	1.0861	2.16%	6.41%	0.1815	1.0000
0.25	3.39%	9.18%	0.2610	1.0000	3.39%	9.18%	0.2610	1.0000
0.50	5.52%	15.60%	0.2895	1.0000	5.52%	15.60%	0.2895	1.0000
0.75	7.64%	22.56%	0.2944	1.0000	7.64%	22.56%	0.2944	1.0000
1.00	9.76%	29.67%	0.2954	1.1119	9.69%	29.43%	0.2951	1.0000

Table 20: Comparison of portfolio statistics between 2(b) and 3(c).

This also explains the differences in the weights. For $\gamma = 0.25$, 0.50 and 0.75, the weights are identical because the solution of 2(b) already satisfies $\mathcal{L}(x) = 1$, so the constraint is not binding. For the other values of γ, the weights differ because the leverage constraint limits the long/short positions.

These results are consistent with the relationship observed in point 2(d), where the leverage ratio $\mathcal{L}(x^\star(\gamma))$ is plotted as a function of the risk tolerance parameter γ. As shown in Figure 2, leverage reaches its minimum value equal to 1 for intermediate values of γ, while it increases for low or high levels of risk tolerance. This explains why the leverage constraint is binding only for some values of γ, while for others the unconstrained solution already satisfies $\mathcal{L}(x) = 1$.

1.4 Point 4

We consider long-only MVO portfolios such that $\sum_{i=1}^{n} x_i = 1$ and $0 \leq x_i \leq 1$.

(a) Using the γ-problem, find the optimal solution $x^\star(\gamma)$ when the coefficient γ is equal to 0, 0.10, 0.20, 0.50 and 1.00. For each optimized portfolio, compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$ and its Sharpe ratio $SR(x^\star(\gamma) \mid r)$. Compare the two solutions 3(c) and 4(a). Comments on these results.

We solve the γ-problem under the long-only constraints

$$\sum_{i=1}^{8} x_i = 1, \qquad 0 \leq x_i \leq 1,$$

for $\gamma \in \{0, 0.10, 0.20, 0.25, 0.50, 1.00\}$. For each optimal portfolio we compute

$$\mu_p = \mu^\top x^\star(\gamma), \qquad \sigma_p = \sqrt{x^\star(\gamma)^\top \Sigma x^\star(\gamma)}, \qquad SR_p = \frac{\mu_p - r}{\sigma_p}.$$

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	μ_p	σ_p	SR
0.00	25.03%	15.60%	25.92%	15.63%	14.93%	2.89%	0.00%	0.00%	1.35%	5.74%	0.0611
0.10	20.91%	13.57%	24.20%	15.64%	18.57%	4.95%	2.15%	0.00%	2.16%	6.41%	0.1815
0.20	17.01%	11.60%	22.36%	15.44%	21.54%	6.71%	5.32%	0.00%	2.98%	8.11%	0.2447
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%	3.39%	9.18%	0.2610
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%	5.52%	15.60%	0.2895
1.00	0.00%	0.00%	3.86%	8.37%	28.87%	13.41%	20.08%	25.41%	9.69%	29.43%	0.2951

Table 21: Optimal long-only portfolios for point 4(a).

γ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	μ_p	σ_p	SR
0.00	25.03%	15.60%	25.92%	15.63%	14.93%	2.89%	0.00%	0.00%	1.35%	5.74%	0.0611
0.10	20.91%	13.57%	24.20%	15.64%	18.57%	4.95%	2.15%	0.00%	2.16%	6.41%	0.1815
0.25	15.11%	10.64%	21.45%	15.33%	22.96%	7.55%	6.85%	0.11%	3.39%	9.18%	0.2610
0.50	8.31%	6.99%	17.19%	13.92%	25.59%	9.55%	10.97%	7.47%	5.52%	15.60%	0.2895
0.75	1.52%	3.34%	12.94%	12.51%	28.23%	11.56%	15.09%	14.83%	7.64%	22.56%	0.2944
1.00	0.00%	0.00%	3.86%	8.37%	28.87%	13.41%	20.08%	25.41%	9.69%	29.43%	0.2951

Table 22: Optimal long/short portfolios obtained in point 3(c).

The results of the long-only problem (4a) are identical to those obtained in the long/short formulation of point 3(c). This occurs because the optimal portfolios in 3(c) already satisfy $x_i \geq 0$ for all assets. Therefore, the long-only constraint is not binding and both optimization problems lead to the same efficient portfolios.

(b) Compare the efficient frontier by considering granular values of γ with the long/short efficient frontier obtained in Question 2(c). Comment on these results.

To compare the two efficient frontiers, we solve the γ-problem for a fine grid of values of the risk tolerance parameter γ. For each value of γ, the following quadratic optimization problem is solved:

$$x^\star(\gamma) = \arg\min_x \left(\frac{1}{2} x^\top \Sigma x - \gamma x^\top (\mu - r\mathbf{1}) \right)$$

and the corresponding expected return and volatility are computed as

$$\mu_p = \mu^\top x^\star(\gamma), \qquad \sigma_p = \sqrt{x^\star(\gamma)^\top \Sigma x^\star(\gamma)}.$$

Two versions of the problem are considered:

- **Long/short frontier (Question 2(c)):**

$$\sum_{i=1}^{n} x_i = 1, \qquad -10 \le x_i \le 10$$

- **Long-only frontier (Question 4):**

$$\sum_{i=1}^{n} x_i = 1, \qquad 0 \le x_i \le 1$$

By solving the optimization problem for many values of γ, we obtain a dense set of portfolios that traces the efficient frontier in the (σ, μ) plane.

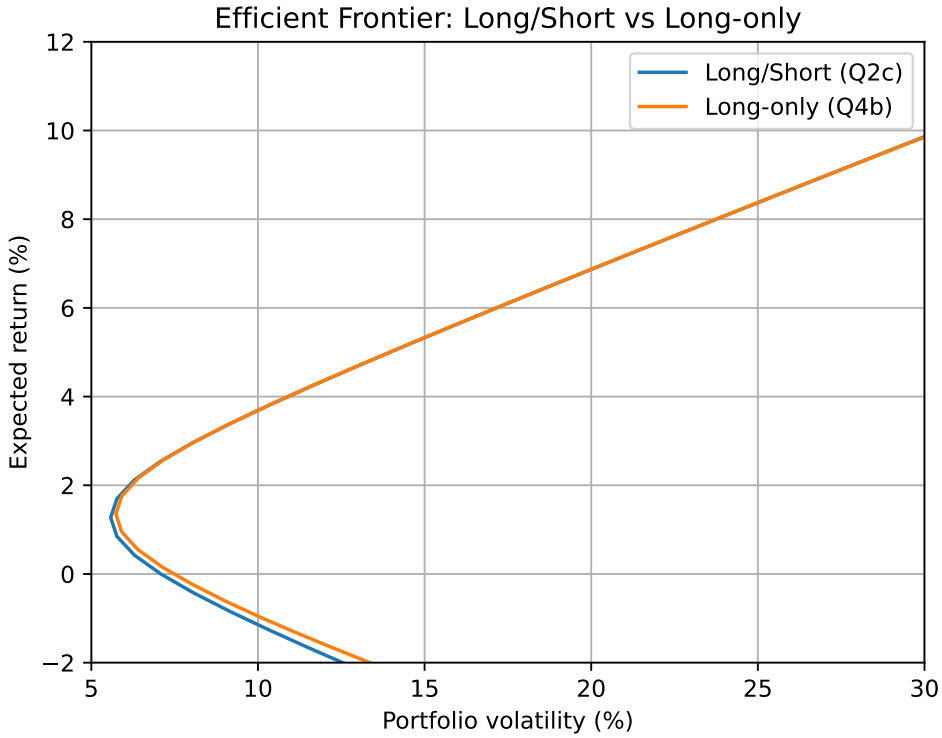


Figure 4: Efficient frontier comparison: long/short vs long-only portfolios.

The long/short frontier is slightly more efficient than the long-only frontier. This result is consistent with the theory of constrained portfolio optimization: allowing short positions enlarges the feasible set of portfolios and enables the optimizer to better exploit the covariance structure of asset returns. In particular, short-selling allows the portfolio to hedge exposures between negatively correlated assets and therefore achieve a lower variance for a given level of expected return.

When the constraint $x_i \geq 0$ is imposed, the feasible set becomes smaller and some optimal unconstrained portfolios are no longer attainable. As a consequence, the efficient frontier obtained under long-only constraints is shifted slightly inward, meaning that for the same level of expected return the portfolio volatility is marginally higher.

In this dataset, however, the difference between the two frontiers is relatively small. This indicates that the optimal portfolios of the unconstrained problem involve only limited short positions, so that imposing the long-only constraint does not dramatically affect the efficient frontier.

(c) Using a bisection algorithm, calculate the MVO portfolio if we target an ex-ante volatility of 10% and 20%. Give the corresponding value of γ of the QP problem. For each optimized portfolio, compute its expected return $\mu(x^\star(\gamma))$, its volatility $\sigma(x^\star(\gamma))$ and its Sharpe ratio $SR(x^\star(\gamma) \mid r)$.

We use a bisection algorithm on γ to find $\gamma^\star$ such that the optimal solution $x^\star(\gamma^\star)$ satisfies the target volatility $\sigma(x^\star(\gamma^\star))$. The algorithm iteratively halves the interval of possible values of γ and selects the subinterval whose portfolio volatility is closer to the target until convergence.

Target σ	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
10%	14.16%	10.13%	20.86%	15.14%	23.32%	7.83%	7.43%	1.13%
20%	3.99%	4.66%	14.48%	13.02%	27.27%	10.83%	13.59%	12.15%

Table 23: Optimal long-only portfolio weights corresponding to the target volatility levels.

Target σ	$\gamma^\star$	$\mu(x^\star(\gamma^\star))$	$\sigma(x^\star(\gamma^\star))$	$SR(x^\star(\gamma^\star) \mid r)$
10%	0.2848	3.69%	10.00%	0.2690
20%	0.6591	6.87%	20.00%	0.2934

Table 24: Optimal long-only portfolio statistics for the considered target volatility levels.

As expected, the portfolio targeting 20% volatility is associated with a higher value of $\gamma^\star$ than the portfolio targeting 10% volatility. This higher risk tolerance leads to a higher expected return and a higher Sharpe ratio.

(d) Find the long-only tangency portfolio $x^\star_{\text{MSR}}$. Compute its expected return $\mu(x^\star_{\text{MSR}})$, its volatility $\sigma(x^\star_{\text{MSR}})$ and its Sharpe ratio $SR(x^\star_{\text{MSR}}|r)$. Compare these results with those obtained in the long/short case.

The long-only efficient frontier is obtained by solving the mean-variance optimization problem

$$\min_x \left(\frac{1}{2} x^\top \Sigma x - \gamma x^\top (\mu - r\mathbf{1}) \right)$$

subject to the long-only constraints

$$\sum_{i=1}^{n} x_i = 1, \qquad 0 \leq x_i \leq 1.$$

Since no analytical solution is available under inequality constraints, the problem is solved numerically for a grid of values of γ, generating the long-only efficient frontier. The tangency portfolio is then approximated by selecting, among these portfolios, the one that maximizes the Sharpe ratio.

The resulting long-only tangency portfolio is reported and compared in Table 25 and Table 26.

Case	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
LS analytical (2f)	-9.21%	-2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%
LS brute force (2g)	-9.21%	-2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%
Extended (2h)	-9.21%	-2.43%	6.22%	10.28%	32.39%	14.72%	21.59%	26.44%
Long-only MSR (4d)	0.00%	0.00%	5.57%	9.21%	29.01%	13.18%	19.34%	23.69%

Table 25: Portfolio weights for the different tangency portfolio computations.

Case	γ	μ	σ	SR	L
LS analytical (2f)	1.1447	10.99%	33.82%	0.2954	1.2329
LS brute force (2g)	1.1447	10.99%	33.82%	0.2954	1.2329
Extended (2h)	1.1447	10.99%	33.82%	0.2954	1.2329
Long-only MSR (4d)	0.9660	9.42%	28.51%	0.2952	1.0000

Table 26: Statistics of the tangency portfolios obtained with different methods.

Compared with the long/short case, the long-only tangency portfolio exhibits a lower expected return and a lower volatility, with a slightly smaller Sharpe ratio. This difference is due to the long-only constraint $x_i \geq 0$, which restricts the feasible set of portfolios. In the long/short case the optimizer can exploit negative weights to hedge exposures between assets, which allows a slightly higher Sharpe ratio. When short-selling is not allowed, the optimization problem becomes more constrained and the resulting tangency portfolio is therefore slightly less efficient.

(e) Compute the beta coefficient β_i of each asset with respect to the long-only tangency portfolio $x^\star_{\text{MSR}}$. Deduce the implied expected return μ_i that is priced in by the market, and the corresponding alpha coefficient α_i of each asset.

We assume that the market portfolio is the long-only tangency portfolio $x^\star_{MSR}$ obtained in the previous question. According to the CAPM framework, the beta of asset i with respect to the market portfolio is defined as

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \frac{(\Sigma x^\star_{MSR})_i}{x^{\star\top}_{MSR}\Sigma x^\star_{MSR}},$$

where R_m denotes the return of the market portfolio.

Given the beta coefficient, the CAPM implies that the expected return priced by the market is

$$\mu_i^{CAPM} = r + \beta_i \left(\mu(x^\star_{MSR}) - r\right).$$

Finally, the alpha coefficient of asset i is defined as the difference between its actual expected return and the return implied by the CAPM:

$$\alpha_i = \mu_i - \mu_i^{CAPM}.$$

The results obtained for each asset are reported in Table 27.

Asset	β_i	μ_i	μ_i^{CAPM}	α_i
1	-0.6901	-5.00%	-4.81%	-0.19%
2	-0.2070	-0.80%	-0.74%	-0.06%
3	0.2139	2.80%	2.80%	0.00%
4	0.4278	4.60%	4.60%	0.00%
5	0.7130	7.00%	7.00%	0.00%
6	0.8556	8.20%	8.20%	0.00%
7	1.0695	10.00%	10.00%	0.00%
8	1.7824	16.00%	16.00%	0.00%

Table 27: Beta coefficients, CAPM implied returns and alpha values.

We observe that the alpha coefficients are very close to zero. This indicates that the expected returns of the assets are almost perfectly explained by the CAPM relationship with respect to the long-only tangency portfolio.

1.5 Point 5

Throughout this point we use portfolio volatility as the risk measure,

$$R(x) = \sigma(x) = \sqrt{x^\top \Sigma x}.$$

The associated Euler risk decomposition is given by

$$MR_i(x) = \frac{(\Sigma x)_i}{\sigma(x)}, \qquad RC_i(x) = x_i\, MR_i(x), \qquad RC_i^\star(x) = \frac{RC_i(x)}{\sigma(x)}.$$

Hence, the total portfolio risk satisfies

$$\sum_{i=1}^{n} RC_i(x) = \sigma(x), \qquad \sum_{i=1}^{n} RC_i^\star(x) = 1.$$

(a) Give the risk decomposition of the long-only tangency portfolio $x_{MSR}^\star$ (MSR).

We consider the long-only tangency portfolio $x_{MSR}^\star$ obtained previously.

Table 28: Risk decomposition of the long-only tangency portfolio $x^\star_{MSR}$

Asset	x_i	MR_i	RC_i	$RC_i^\star$
1	0.00%	-19.68%	0.00%	0.00%
2	0.00%	-5.90%	0.00%	0.00%
3	5.57%	6.10%	0.34%	1.19%
4	9.21%	12.20%	1.12%	3.94%
5	29.01%	20.33%	5.90%	20.69%
6	13.18%	24.39%	3.22%	11.28%
7	19.34%	30.49%	5.90%	20.69%
8	23.69%	50.82%	12.04%	42.22%
$R(x)$	28.51%			

The long-only tangency portfolio is highly concentrated in the last assets, especially x_8, which alone contributes more than 42% of total portfolio risk. By contrast, the first two assets have essentially zero weight and therefore zero risk contribution.

(b) Give the risk decomposition of the equally-weighted portfolio (EW).

The equally-weighted portfolio is defined by:

$$x_{EW} = \left(\frac{1}{n}, \ldots, \frac{1}{n}\right).$$

Table 29: Risk decomposition of the equally-weighted portfolio x_{EW}

Asset	x_i	MR_i	RC_i	$RC_i^\star$
1	12.50%	-17.26%	-2.16%	-14.22%
2	12.50%	-3.63%	-0.45%	-2.99%
3	12.50%	6.56%	0.82%	5.40%
4	12.50%	12.47%	1.56%	10.27%
5	12.50%	19.64%	2.46%	16.18%
6	12.50%	24.33%	3.04%	20.05%
7	12.50%	29.86%	3.73%	24.60%
8	12.50%	49.40%	6.18%	40.70%
$R(x)$	15.17%			

Although the EW portfolio is perfectly diversified in terms of weights, it is not equally diversified in terms of risk. In particular, assets x_1 and x_2 have negative marginal and total risk contributions, whereas asset x_8 represents the largest share of total risk.

(c) Give the risk decomposition of the long-only minimum variance portfolio (MV).

The long-only minimum variance portfolio is obtained by solving

$$x_{MV} = \arg\min_x \frac{1}{2} x^\top \Sigma x$$

subject to

$$\mathbf{1}^\top x = 1, \qquad 0 \le x \le \mathbf{1}.$$

In our implementation this corresponds to setting $\gamma = 0$ in the quadratic programming problem used previously.

Table 30: Risk decomposition of the minimum variance portfolio x_{MV}

Asset	x_i	MR_i	RC_i	$RC_i^\star$
1	25.03%	5.74%	1.44%	25.03%
2	15.60%	5.74%	0.90%	15.60%
3	25.92%	5.74%	1.49%	25.92%
4	15.63%	5.74%	0.90%	15.63%
5	14.93%	5.74%	0.86%	14.93%
6	2.89%	5.74%	0.17%	2.89%
7	0.00%	6.11%	0.00%	0.00%
8	0.00%	10.18%	0.00%	0.00%
$R(x)$	5.74%			

(d) Give the risk decomposition of the long-only most diversified portfolio (MDP).

The most diversified portfolio maximizes the diversification ratio

$$DR(x) = \frac{x^\top \sigma}{\sqrt{x^\top \Sigma x}},$$

where $\sigma = (\sigma_1, \ldots, \sigma_n)^\top$ is the vector of individual volatilities. The optimization problem is

$$x_{MDP} = \arg\max_x \ln DR(x)$$
$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top x = 1 \\ 0 \le x \le \mathbf{1} \end{cases}$$

Table 31: Risk decomposition of the most diversified portfolio x_{MDP}

Asset	x_i	MR_i	RC_i	$RC_i^\star$
1	35.96%	7.08%	2.55%	38.58%
2	16.49%	4.84%	0.80%	12.09%
3	11.55%	3.30%	0.38%	5.78%
4	8.22%	4.61%	0.38%	5.74%
5	9.53%	6.10%	0.58%	8.80%
6	5.94%	7.73%	0.46%	6.95%
7	6.36%	9.15%	0.58%	8.81%
8	5.95%	14.70%	0.87%	13.24%
$R(x)$	6.60%			

(e) Compute the equal risk contribution portfolio using the CCD algorithm. Give its risk decomposition.

The equal risk contribution (ERC) portfolio equalizes the contribution of each asset to total portfolio risk:

$$RC_i(x) = RC_j(x), \quad \forall i, j.$$

The portfolio is computed using the cyclical coordinate descent (CCD) algorithm.

Table 32: Risk decomposition of the equal risk contribution portfolio x_{ERC}

Asset	x_i	MR_i	RC_i	$RC_i^\star$
1	29.69%	2.66%	0.79%	12.50%
2	17.20%	4.59%	0.79%	12.50%
3	16.07%	4.92%	0.79%	12.50%
4	11.40%	6.93%	0.79%	12.50%
5	8.96%	8.82%	0.79%	12.50%
6	6.96%	11.35%	0.79%	12.50%
7	5.97%	13.23%	0.79%	12.50%
8	3.77%	20.95%	0.79%	12.50%
$R(x)$	6.32%			

As expected, each asset contributes the same fraction of portfolio risk (12.5%). This confirms that the CCD algorithm successfully identifies the ERC allocation.

(f) Compute the beta $\beta(x \mid b)$ of the portfolios MSR, EW, MV, MDP and ERC with respect to the benchmark b when b is the long-only tangency portfolio $x_{MSR}^\star$. Same question when b is the EW portfolio. Comment on these results.

For a given benchmark b, the beta of portfolio x is defined by

$$\beta(x \mid b) = \frac{x^\top \Sigma b}{b^\top \Sigma b}.$$

We compute this quantity for the five portfolios (MSR, EW, MV, MDP, ERC), first with benchmark $b = x_{MSR}^\star$, and then with benchmark $b = x_{EW}$.

Table 33: Portfolio betas with respect to the benchmark $b = x_{MSR}^\star$

Portfolio	$\beta(x \mid x_{MSR}^\star)$
MSR	1.0000
EW	0.5206
MV	0.0485
MDP	0.0702
ERC	0.0971

Table 34: Portfolio betas with respect to the benchmark $b = x_{EW}$

Portfolio	$\beta(x \mid x_{EW})$
MSR	1.8387
EW	1.0000
MV	0.1581
MDP	0.2062
ERC	0.2520

When the benchmark is the long-only tangency portfolio, the MSR portfolio has beta equal to one by construction. All the other portfolios have betas well below one, which means that they are much more defensive than the tangency portfolio. In particular, the MV, MDP and ERC portfolios exhibit very low exposure to the systematic risk of the MSR benchmark.

When the benchmark is the equally-weighted portfolio, the EW portfolio has beta equal to one, while the MSR portfolio has a beta significantly above one (1.8456). This confirms that the tangency portfolio is much more aggressive than the equally-weighted allocation. By contrast, the MV, MDP and ERC portfolios again have betas below one, which highlights their more conservative risk profile.

Overall, these results are consistent with the nature of the portfolios: the MSR portfolio is the most aggressive one, the EW portfolio has an intermediate exposure, whereas MV, MDP and ERC are substantially more defensive with respect to both benchmarks.

This ranking is also consistent with the theoretical ordering of portfolio volatilities, which states that

$$\sigma(x_{MV}) \leq \sigma(x_{ERC}) \leq \sigma(x_{EW}).$$

Since beta reflects the sensitivity to the benchmark and is closely related to the portfolio risk, the lower betas observed for MV and ERC portfolios are coherent with their lower volatility levels, while the higher beta of the MSR portfolio reflects its more aggressive risk profile.

2 Equity portfolio optimization with ESG and climate risk objectives

We consider an investment universe of $n = 8$ stocks with two sectors ($Sector_1$ and $Sector_2$). The expected return μ_i and the volatility σ_i of each stock i are reported below:

Table 35: Financial and climate metrics of the investment universe

Stock i	Sector	b_i	μ_i	σ_i	S_i	CI_i	CM_i	GI_i
1	$Sector_1$	9.50%	5.00%	20.0%	−2.0	80	−5.0%	5.0%
2	$Sector_2$	15.50%	5.50%	22.0%	+2.5	200	−7.5%	80.5%
3	$Sector_1$	5.50%	6.00%	25.0%	+1.5	390	−1.5%	15.0%
4	$Sector_1$	8.50%	4.00%	18.0%	+2.0	800	−2.0%	0.0%
5	$Sector_2$	10.00%	7.00%	45.0%	−1.0	60	+8.0%	2.0%
6	$Sector_2$	25.00%	10.00%	80.0%	−0.5	120	−4.0%	0.0%
7	$Sector_2$	17.00%	8.75%	35.0%	−0.5	135	−7.0%	60.0%
8	$Sector_1$	9.00%	6.25%	40.5%	+0.5	580	+2.0%	20.0%

The correlation matrix is equal to:

$$C = (\rho_{i,j}) = \begin{pmatrix} 100\% & & & & & & & \\ 50\% & 100\% & & & & & & \\ 30\% & 30\% & 100\% & & & & & \\ 60\% & 60\% & 60\% & 100\% & & & & \\ 40\% & 30\% & 50\% & 30\% & 100\% & & & \\ 30\% & 20\% & 40\% & 70\% & 50\% & 100\% & & \\ 40\% & 60\% & 50\% & 60\% & 50\% & 60\% & 100\% & \\ 30\% & 30\% & 50\% & 30\% & 30\% & 30\% & 60\% & 100\% \end{pmatrix}$$

In Table 1, we report, for each stock, its weight b_i in the benchmark, the corresponding ESG score S_i, the Scope 1+2 carbon intensity CI_i in $tCO_2e/\$$ mn, the carbon momentum CM_i, and the green intensity GI_i measured as the ratio of green capex to total capex over the past three years.

2.1 Point 1

(a) Compute the covariance matrix Σ.

The covariance matrix Σ is obtained from the volatilities σ_i and the correlation matrix $C = (\rho_{ij})$ using the standard relationship between covariance and correlation:

$$\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$$

In matrix form:

$$\Sigma = \operatorname{diag}(\sigma)\, C \operatorname{diag}(\sigma)$$

where $\operatorname{diag}(\sigma)$ is the diagonal matrix whose diagonal is the vector of volatilities.
The resulting covariance matrix is:

$$\Sigma = \begin{pmatrix} 4.00\% & 2.20\% & 1.50\% & 2.16\% & 3.60\% & 4.80\% & 2.80\% & 2.43\% \\ 2.20\% & 4.84\% & 1.65\% & 2.38\% & 2.97\% & 3.52\% & 4.62\% & 2.67\% \\ 1.50\% & 1.65\% & 6.25\% & 2.70\% & 5.62\% & 8.00\% & 4.38\% & 5.06\% \\ 2.16\% & 2.38\% & 2.70\% & 3.24\% & 2.43\% & 10.08\% & 3.78\% & 2.19\% \\ 3.60\% & 2.97\% & 5.62\% & 2.43\% & 20.25\% & 18.00\% & 7.88\% & 5.47\% \\ 4.80\% & 3.52\% & 8.00\% & 10.08\% & 18.00\% & 64.00\% & 16.80\% & 9.72\% \\ 2.80\% & 4.62\% & 4.38\% & 3.78\% & 7.88\% & 16.80\% & 12.25\% & 8.50\% \\ 2.43\% & 2.67\% & 5.06\% & 2.19\% & 5.47\% & 9.72\% & 8.50\% & 16.40\% \end{pmatrix}$$

(b) Compute the expected return $\mu(b)$, the volatility $\sigma(b)$ and the Sharpe ratio $SR(b \mid r)$ of the benchmark.

Given b the vector of benchmark weights, the expected return of the benchmark portfolio b is computed as the weighted average of the individual expected returns:

$$\mu(b) = b^\top \mu$$

where μ is the vector of expected returns, the volatility of the benchmark is computed using the covariance matrix Σ:

$$\sigma(b) = \sqrt{b^\top \Sigma b}$$

and finally, the Sharpe ratio of the benchmark given the risk-free rate r is:

$$SR(b|r) = \frac{\mu(b) - r}{\sigma(b)}$$

Using $r = 1\%$, the results are:

$\mu(b)$	$\sigma(b)$	$SR(b \mid r)$
7.25%	32.63%	0.1915

(c) We assume that the benchmark b is the optimal market portfolio. Compute the implied expected returns

Assuming that the benchmark portfolio b is the optimal market portfolio, the implied expected returns are computed as:

$$\tilde{\mu}^{(b)} = r + SR(b \mid r) \cdot \frac{\Sigma b}{\sigma(b)}$$

where Σ is the covariance matrix, b is the benchmark weight vector, $SR(b \mid r)$ is the Sharpe ratio of the benchmark, and $\sigma(b)$ is the volatility of the benchmark.

The resulting implied expected returns are:

Asset	$\tilde{\mu}^{(b)}$
1	2.90%
2	3.03%
3	3.78%
4	3.70%
5	6.68%
6	14.99%
7	6.50%
8	5.11%

We recall that the benchmark portfolio implied return is $\mu(b) = 7.25\%$

(d) Compute the betas $\beta_i(b)$ of each stock with respect to the benchmark b and deduce the CAPM expected returns. Compare the results of (c) and (d) and comment on them.

The beta of stock i with respect to the benchmark portfolio b is computed as:

$$\beta_i(b) = \frac{\mathrm{Cov}(R_i, R_b)}{\mathrm{Var}(R_b)} = \frac{(\Sigma b)_i}{b^\top \Sigma b}$$

Using the CAPM relation, the expected return of each stock is:

$$\tilde{\mu}_i^{(\mathrm{capm})} = r + \beta_i(b)\big(\mu(b) - r\big)$$

The results, together with the implied expected returns from point (c), are:

Asset	$\beta_i(b)$	$\tilde{\mu}_i^{(b)}$	$\tilde{\mu}_i^{(CAPM)}$	Δ_i
1	0.3045	2.90%	2.90%	0.00%
2	0.3246	3.03%	3.03%	0.00%
3	0.4447	3.78%	3.78%	0.00%
4	0.4321	3.70%	3.70%	0.00%
5	0.9088	6.68%	6.68%	0.00%
6	2.2386	14.99%	14.99%	0.00%
7	0.8811	6.50%	6.50%	0.00%
8	0.6584	5.11%	5.11%	0.00%

Comparing the results reported in the table with those obtained in point (c), we observe that the CAPM expected returns equals to the implied expected returns for every asset. Indeed, from point (c) we obtained the implied expected returns

$$\tilde{\mu}^{(b)} = r + SR(b \mid r)\,\frac{\Sigma b}{\sigma(b)}$$

and since

$$SR(b \mid r) = \frac{\mu(b) - r}{\sigma(b)},$$

it follows that

$$\tilde{\mu}_i^{(b)} = r + \frac{(\mu(b) - r)}{\sigma(b)} \frac{(\Sigma b)_i}{\sigma(b)} = r + \frac{(\Sigma b)_i}{\sigma^2(b)} (\mu(b) - r) = r + \beta_i(b)\big(\mu(b) - r\big).$$

The CAPM formula is:

$$\tilde{\mu}_i^{(\text{capm})} = r + \beta_i(b)\big(\mu(b) - r\big) = \tilde{\mu}_i^{(b)}.$$

Then assuming the market portfolio as the benchmark, the implied expected returns in (c) coincide precisely with the CAPM-implied returns in (d).

(e) Deduce the alpha of each stock. Why is $\alpha_i \neq 0$?

The alpha of stock i is defined as the difference between the given expected return and the implied expected return obtained in point (c):

$$\alpha_i = \mu_i - \tilde{\mu}_i^{(b)}$$

The resulting values are:

Asset	α_i
1	2.10%
2	2.47%
3	2.22%
4	0.30%
5	0.32%
6	-4.99%
7	2.25%
8	1.14%

The values α_i are not equal to zero because the expected returns μ_i given in the data do not coincide with the implied expected returns $\tilde{\mu}_i^{(b)}$ obtained under the assumption that the benchmark portfolio b is the market portfolio.

If the given returns were fully consistent with the CAPM equilibrium associated with b, we would have $\mu_i = \tilde{\mu}_i^{(b)}$ for every asset and therefore $\alpha_i = 0$. Since this is not the case, the alphas are non-zero.

(f) Compute $CI(b)$, $CM(b)$, $GI(b)$, and the ESG score $S(b)$.

The ESG characteristics of the benchmark portfolio b are obtained as the weighted averages of the corresponding metrics of the individual stocks. For a generic metric X_i, the portfolio value is computed as:

$$X(b) = \sum_{i=1}^{n} b_i X_i = b^\top X$$

Applying this formula to the carbon intensity (CI), carbon momentum (CM), green intensity (GI), and ESG score S, we obtain:

$$CI(b) = b^\top CI$$
$$CM(b) = b^\top CM$$
$$GI(b) = b^\top GI$$
$$S(b) = b^\top S$$

The resulting ESG metrics of the benchmark are:

Metric	Value
CI(b)	239.2000
CM(b)	−3.10%
GI(b)	25.98%
S(b)	0.1850

(g) Compute the carbon intensity, the carbon momentum, the green intensity and the ESG score for each sector

The ESG metrics for each sector are computed as the weighted averages of the corresponding stock metrics within that sector. Since the benchmark weights b_i do not sum to one within each sector, they are first normalized:

$$\tilde{b}_i = \frac{b_i}{\sum_{j \in s} b_j}, \qquad i \in s$$

where s denotes the set of stocks belonging to the considered sector.

The sector-level metric is then obtained as:

$$X(s) = \sum_{i \in s} \tilde{b}_i X_i$$

where X_i represents the metric of stock i (carbon intensity CI, carbon momentum CM, green intensity GI, or ESG score S).

The resulting sector metrics are:

Sector	**CI**	**CM**	**GI**	**S**
Sector_1	459.23	-1.68%	9.54%	0.331
Sector_2	133.26	-3.78%	33.89%	0.115

2.2 Point 2

The investor's objective is to minimize the volatility of the tracking error relative to the benchmark while incorporating ESG and climate risk constraints. At each step, the investor adds a new constraint, resulting in the accumulation of constraints.

(a) The investor begins by adding a decarbonization constraint: $CI(w) \leq (1 - 30\%)\, CI(b)$. Give the QP problem and find the optimal solution.

The optimization problem can first be written in the general quadratic form subject to linear equality and inequality constraints:

$$w^\star = \arg\min_w \left(\frac{1}{2} w^\top Q w - w^\top R \right)$$

$$\text{s.t.} \quad \begin{cases} Aw = B \\ Cw \leq D \\ w^- \leq x \leq w^+ \end{cases}$$

In this case, the objective function to minimize is the tracking-error variance with respect to the benchmark b,

$$\frac{1}{2}(w-b)^\top \Sigma (w-b) = \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w + \frac{1}{2} b^\top \Sigma b$$

Since the last term is costant, in our case the minimization problem can be written as:

$$\min_w \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$

$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1 \\ 0 \leq w_i \leq 1, \quad i = 1, \ldots, n \\ w^\top CI \leq (1 - 30\%)\, CI(b) \end{cases}$$

which, since $(\Sigma b)^\top w = w^\top (\Sigma b)$, corresponds to the general quadratic form setting:

$$\begin{aligned} Q &= \Sigma, & R &= \Sigma b \\ A &= \mathbf{1}^\top, & B &= 1 \\ C &= CI^\top, & D &= (1 - 0.3)\, CI(b) \\ x^- &= 0, & x^+ &= 1 \end{aligned}$$

Since $CI(b) = 239.2$, the carbon constraint becomes:

$$w^\top CI \leq 0.7 \times 239.2 = 167.44.$$

The optimal solution is:

Asset	$w_i^\star$
1	16.62%
2	16.28%
3	6.45%
4	0.00%
5	9.21%
6	25.77%
7	19.98%
8	5.69%

The corresponding portfolio carbon intensity is:

Metric	Value	Variation
$CI(w^\star)$	167.4400	-30.00%

The important point is that the decarbonization constraint is binding at the optimum, since the portfolio reaches exactly the maximum admissible carbon intensity level.

(b) The investor adds a green intensity constraint: $GI(w) \geq (1 + 50\%)\, GI(b)$. What does this constraint mean? Give the QP problem and find the optimal solution.

The constraint

$$GI(w) \geq (1 + 50\%)\, GI(b)$$

means that the portfolio must have a green intensity at least 50% higher than the benchmark. In other words, the portfolio must increase its exposure to assets with higher green intensity in order to raise the weighted average $GI(w)$ above the benchmark level.

The optimization problem is to minimize the tracking-error variance with respect to the benchmark under the decarbonization and green-intensity constraints:

$$\min_{w} \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$

$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1, \\ 0 \leq w_i \leq 1, \qquad i = 1, \ldots, n, \\ w^\top CI \leq (1 - 30\%)\, CI(b). \\ w^\top GI \geq (1 + 50\%)\, GI(b). \end{cases}$$

Therefore, the corresponding QP problem in the general form is:

$$\min_{w} \quad \frac{1}{2} w^\top Q w - R^\top w$$

$$\text{s.t.} \quad \begin{cases} Aw = B \\ Cw \leq D \\ w^- \leq x \leq w^+ \end{cases}$$

where:

$$Q = \Sigma, \qquad R = \Sigma b$$

$$A = \mathbf{1}^\top, \qquad B = 1$$

$$C = \begin{bmatrix} CI^\top \\ -GI^\top \end{bmatrix}, \quad D = \begin{bmatrix} (1 - 0.3)\, CI(b) \\ -(1 + 0.5)\, GI(b) \end{bmatrix}$$

$$x^- = 0, \qquad x^+ = 1$$

Since $CI(b) = 239.2$ and $GI(b) = 25.9775\%$, the two constraints become:

$$w^\top CI \leq 167.44, \qquad w^\top GI \geq 38.97\%.$$

The optimal solution is:

Asset	$w_i^\star$
1	6.38%
2	29.82%
3	3.20%
4	0.00%
5	8.75%
6	25.72%
7	21.90%
8	4.23%

The corresponding portfolio metrics are:

Metric	Value	Variation
$CI(w^\star)$	167.4400	−30.00%
$GI(w^\star)$	38.97%	50.00%

An important point is that both constraints are binding at the optimum, since the optimal portfolio reaches exactly the carbon-intensity target and the green-intensity target.

(c) The investor adds a third constraint: $CM(w) \leq (1+50\%)\, CM(b)$. What does this constraint mean? Give the QP problem and find the optimal solution.

The constraint

$$CM(w) \leq (1 + 50\%)\, CM(b)$$

imposes a restriction on the portfolio's carbon momentum relative to the benchmark. Carbon momentum measures the trend in firms' carbon emissions over time: negative values indicate decreasing emissions, while higher (or less negative) values correspond to a worse emissions trajectory.

Since assets' carbon momentum values are negative, this constraint implies that the portfolio's carbon momentum must be at least 1.5 times lower than the benchmark's one. Therefore, the portfolio's carbon momentum must be even more negative than the benchmark's, meaning that the portfolio exhibits a better emissions trend. The corresponding portfolio optimization problem is

$$\min_w \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$
$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1, \\ 0 \leq w_i \leq 1, \quad i = 1, \ldots, n, \\ w^\top CI \leq (1 - 30\%)\, CI(b). \\ w^\top GI \geq (1 + 50\%)\, GI(b). \\ w^\top CM \leq (1 + 50\%)\, CM(b) \end{cases}$$

that can be written in the standard QP form as

$$\min_{w} \quad \frac{1}{2} w^\top Q w - R^\top w$$
$$\text{s.t.} \quad \begin{cases} Aw = B \\ Cw \leq D \\ w^- \leq x \leq w^+ \end{cases}$$

where:

$$Q = \Sigma, \qquad R = \Sigma b$$
$$A = \mathbf{1}^\top, \qquad B = 1$$
$$C = \begin{bmatrix} CI^\top \\ -GI^\top \\ CM^\top \end{bmatrix}, \quad D = \begin{bmatrix} (1-0.3)\, CI(b) \\ -(1+0.5)\, GI(b) \\ (1+0.5)\, CM(b) \end{bmatrix}$$
$$x^- = 0, \qquad x^+ = 1$$

Since $CI(b) = 239.2$, $GI(b) = 25.9775\%$ and $CM(b) = -3.10\%$, the constraints become

$$w^\top CI \leq 167.44, \qquad w^\top GI \geq 38.97\%, \qquad w^\top CM \leq -4.65\%.$$

The optimal portfolio is

Asset	$w_i^\star$
1	7.63%
2	29.08%
3	4.18%
4	0.00%
5	6.74%
6	25.95%
7	22.82%
8	3.60%

The corresponding portfolio metrics are:

Metric	Value	Variation
$CI(w^\star)$	167.4400	−30.00%
$GI(w^\star)$	38.97%	50.00%
$CM(w^\star)$	−4.65%	50.00%

The three ESG constraints are binding at the optimum, meaning that the optimal portfolio satisfies each constraint.

(d) The investor adds a fourth constraint: $S(w) \geq S(b) + 0.5$. What does this constraint mean? Give the QP problem and find the optimal solution.

The constraint

$$S(w) \geq S(b) + 0.5$$

requires the portfolio ESG score to be at least 0.5 points higher than the benchmark ESG score.

The optimization problem is

$$\min_{w} \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$

$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1, \\ 0 \leq w_i \leq 1, \quad i = 1, \ldots, n, \\ w^\top CI \leq (1 - 30\%)\, CI(b), \\ w^\top GI \geq (1 + 50\%)\, GI(b), \\ w^\top CM \leq (1 + 50\%)\, CM(b), \\ w^\top S \geq S(b) + 0.5. \end{cases}$$

that can be written in standard QP form as

$$\min_{w} \quad \frac{1}{2} w^\top Q w - R^\top w$$

$$\text{s.t.} \quad \begin{cases} Aw = B \\ Cw \leq D \\ w^- \leq x \leq w^+ \end{cases}$$

where:

$$Q = \Sigma, \qquad R = \Sigma b$$

$$A = \mathbf{1}^\top, \qquad B = 1$$

$$C = \begin{bmatrix} CI^\top \\ -GI^\top \\ CM^\top \\ -S^\top \end{bmatrix}, \quad D = \begin{bmatrix} (1 - 0.3)\, CI(b) \\ -(1 + 0.5)\, GI(b) \\ (1 + 0.5)\, CM(b) \\ -\big(S(b) + 0.5\big) \end{bmatrix}$$

$$x^- = 0, \qquad x^+ = 1$$

Since $CI(b) = 239.2$, $GI(b) = 25.98\%$, $CM(b) = -3.10\%$ and $S(b) = 0.1850$, the constraints become

$$w^\top CI \leq 167.44, \qquad w^\top GI \geq 38.97\%, \qquad w^\top CM \leq -4.65\%, \qquad w^\top S \geq 0.6850.$$

The optimal portfolio is

Asset	$w_i^\star$
1	2.65%
2	37.56%
3	6.65%
4	0.00%
5	8.26%
6	27.40%
7	16.84%
8	0.64%

The corresponding portfolio metrics are:

Metric	Value	Variation
$CI(w^\star)$	167.4400	-30.00%
$GI(w^\star)$	41.76%	60.75%
$CM(w^\star)$	-4.65%	50.00%
$S(w^\star)$	0.6850	270.27%

The ESG score constraint is binding at the optimum, as the optimal portfolio achieves exactly the required ESG target $S(w^\star) = S(b) + 0.5$..

(e) Compute the tracking error volatility, the carbon intensity, the carbon momentum, the green intensity, the ESG score, and the sector allocation for the optimized portfolios found in 2.(a),2.(b), 2.(c), and 2.(d).

The following table reports the tracking error volatility, carbon intensity (CI), carbon momentum (CM), green intensity (GI), ESG score, and sector allocation for the optimized portfolios obtained in points 2.(a)-2.(d).

Portfolio	TE Vol	CI	CM	GI	S	Sector1	Sector2
2.(a)	1.33%	167.44	-3.73%	28.22%	-0.1209	28.76%	71.24%
2.(b)	3.21%	167.44	-4.38%	38.97%	0.3614	13.82%	86.18%
2.(c)	3.29%	167.44	-4.65%	38.97%	0.3440	15.41%	84.59%
2.(d)	4.41%	167.44	-4.65%	41.76%	0.6850	9.94%	90.06%

Several observations can be made.

First, the carbon intensity is identical across all portfolios ($CI = 167.44$), showing that the decarbonization constraint imposed in 2.(a) remains binding throughout the subsequent optimizations.

Second, the tracking error volatility increases progressively as additional ESG constraints are introduced. This reflects the growing restriction of the feasible set when more sustainability requirements are imposed.

Finally, the sector allocation shifts strongly toward Sector2. As more ESG constraints are added, the optimized portfolios allocate a larger share of weights to this sector, suggesting that Sector2 contains firms with more favorable ESG characteristics.

(f) Compute the implied expected return and the alpha of stock i priced in by each optimized portfolio w.

For a given optimized portfolio w, the Sharpe ratio is

$$SR(w \mid r) = \frac{\mu(w) - r}{\sigma(w)}, \qquad \mu(w) = \mu^\top w, \qquad \sigma(w) = \sqrt{w^\top \Sigma w}.$$

The implied expected return of stock i priced by the portfolio w is

$$\tilde{\mu}_i^{(w)} = r + SR(w \mid r) \cdot \frac{(\Sigma w)_i}{\sigma(w)}.$$

The alpha of stock i with respect to the benchmark equilibrium is defined as

$$\alpha_i(w \mid b) = \tilde{\mu}_i^{(b)} - \tilde{\mu}_i^{(w)}.$$

We now report the implied expected returns and the corresponding alphas for each optimized portfolio obtained in points 2.(a)-2.(d).

Asset	$\tilde{\mu}_i^{(2a)}$	$\alpha_i(2a \mid b)$	$\tilde{\mu}_i^{(2b)}$	$\alpha_i(2b \mid b)$	$\tilde{\mu}_i^{(2c)}$	$\alpha_i(2c \mid b)$	$\tilde{\mu}_i^{(2d)}$	$\alpha_i(2d \mid b)$
1	3.02%	-0.12%	2.93%	-0.03%	2.94%	-0.04%	2.86%	0.05%
2	3.10%	-0.07%	3.35%	-0.32%	3.36%	-0.33%	3.36%	-0.34%
3	3.78%	0.00%	3.70%	0.08%	3.71%	0.07%	3.67%	0.11%
4	3.76%	-0.06%	3.79%	-0.09%	3.83%	-0.13%	3.81%	-0.11%
5	6.87%	-0.19%	6.78%	-0.10%	6.67%	0.01%	6.66%	0.02%
6	15.28%	-0.30%	15.21%	-0.22%	15.30%	-0.31%	15.16%	-0.17%
7	6.66%	-0.15%	6.84%	-0.33%	6.86%	-0.36%	6.61%	-0.11%
8	5.06%	0.05%	4.98%	0.13%	4.98%	0.14%	4.59%	0.53%

First, the alphas are generally small in absolute value, which means that the expected returns implied by the optimized portfolios remain relatively close to those implied by the benchmark equilibrium.

Second, the values of $\tilde{\mu}_i^{(w)}$ and $\alpha_i(w \mid b)$ vary across the four portfolios, reflecting the different sets of ESG and climate constraints imposed in points 2.(a)-2.(d). As the optimization problem becomes more constrained, the composition of the portfolio departs further from the benchmark, and this changes the cross-sectional relation of expected returns.

Overall, these non-zero alphas arise because the optimized portfolios are not the market portfolio anymore. The benchmark-implied returns $\tilde{\mu}_i^{(b)}$ are derived under the assumption that the benchmark is the equilibrium portfolio, whereas the optimized portfolios are obtained under additional decarbonization, green-intensity, carbon-momentum, and ESG-score constraints. As a consequence,the equilibrium expected returns relation does not hold exactly for the constrained portfolios, and the corresponding alphas are not equal to zero.

2.3 Point 3

(a) Compute the carbon intensity of sectors $Sector_1$ and $Sector_2$.

The carbon intensity of each sector is computed as the weighted average of the individual stock carbon intensities within the sector. Since the benchmark weights b_i do not sum to one

inside each sector, they must first be renormalized to 100%. For a given sector s, the normalized benchmark weights are $\tilde{b}_i$ and the corresponding sector carbon intensity is

$$CI(s) = \sum_{i \in s} \tilde{b}_i \, CI_i.$$

Applying this formula to the two sectors gives:

$$CI(Sector_1) = 459.2308,$$
$$CI(Sector_2) = 133.2593.$$

Therefore, $Sector_1$ is substantially more carbon intensive than $Sector_2$.

(b) What is the optimization problem if we impose to reduce the carbon intensity of $Sector_1$ by 30% and the carbon intensity of $Sector_2$ by 50%? Find the optimized portfolio.

The investor minimizes the tracking-error variance relative to the benchmark, subject to long-only and full-investment constraints, while imposing sector-level decarbonization targets.

The sector carbon-intensity constraints are imposed relative to the benchmark sector values computed in point 3.(a). Hence, the targets are

$$CI^{\text{target}}_{Sector_1} = 0.70 \times 459.2308 = 321.4615,$$

$$CI^{\text{target}}_{Sector_2} = 0.50 \times 133.2593 = 66.6296.$$

Since sector carbon intensity is defined with weights renormalized within each sector, the corresponding constraints can be written as weighted-average constraints at the sector level. The QP problem is therefore:

$$\min_{w} \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$
$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1, \\ 0 \le w_i \le 1, \quad i = 1, \dots, n, \\ \displaystyle\sum_{i \in Sector_1} w_i \, (CI_i - 321.4615) \le 0, \\ \displaystyle\sum_{i \in Sector_2} w_i \, (CI_i - 66.6296) \le 0. \end{cases}$$

The optimal portfolio is:

Asset	$w_i^\star$
1	33.52%
2	0.00%
3	3.96%
4	9.21%
5	35.67%
6	4.43%
7	0.00%
8	13.21%

The sector allocation of the optimized portfolio is:

$$w^\star(Sector_1) = 59.90\%, \qquad w^\star(Sector_2) = 40.10\%.$$

The corresponding portfolio characteristics are:

Metric	Value
$TE_{vol}(w^\star \mid b)$	17.33%,
$CI(w^\star)$	219.2644
$CM(w^\star)$	1.02%
$GI(w^\star)$	5.63%
$S(w^\star)$	-0.7396

At the sector level, the constraints are exactly satisfied:

$$CI(Sector_1; w^\star) = 321.4615, \qquad CI(Sector_2; w^\star) = 66.6296.$$

(c) Formulate the QP problem if we also add the sector neutrality constraint. Find the optimized portfolio.

We now add the sector neutrality constraint, which requires the total allocation to each sector to remain equal to the benchmark allocation. Therefore, in addition to the sector decarbonization constraints of point 3.(b), we impose

$$\sum_{i \in Sector_1} w_i = \sum_{i \in Sector_1} b_i = 32.50\%,$$

$$\sum_{i \in Sector_2} w_i = \sum_{i \in Sector_2} b_i = 67.50\%.$$

The resulting quadratic program is

$$\min_{w} \quad \frac{1}{2} w^\top \Sigma w - (\Sigma b)^\top w$$

$$\text{s.t.} \quad \begin{cases} \mathbf{1}^\top w = 1, \\ 0 \le w_i \le 1, \qquad i = 1, \dots, n, \\ \displaystyle\sum_{i \in Sector_1} w_i (CI_i - 321.4615) \le 0, \\ \displaystyle\sum_{i \in Sector_2} w_i (CI_i - 66.6296) \le 0, \\ \displaystyle\sum_{i \in Sector_1} w_i = 0.325, \\ \displaystyle\sum_{i \in Sector_2} w_i = 0.675. \end{cases}$$

The optimal portfolio becomes:

Asset	$w_i^\star$
1	18.98%
2	0.00%
3	0.00%
4	4.95%
5	60.04%
6	7.46%
7	0.00%
8	8.57%

where the sector neutrality constraints are satisfied by construction:

$$w^\star(Sector_1) = 32.50\%, \qquad w^\star(Sector_2) = 67.50\%.$$

The corresponding portfolio characteristics are:

Metric	Value
$TE_{vol}(w^\star \mid b)$	20.32%,
$CI(w^\star)$	149.4500
$CM(w^\star)$	3.63%
$GI(w^\star)$	3.86%
$S(w^\star)$	-0.8755

At the sector level, the carbon-intensity targets are again exactly met:

$$CI(Sector_1; w^\star) = 321.4615, \qquad CI(Sector_2; w^\star) = 66.6296.$$

Compared with point 3.(b), the additional neutrality constraints reduce the investor's freedom to reallocate across sectors. As a result, the tracking error volatility increases, since the feasible set becomes more restrictive.

(d) Compute the tracking error volatility, the carbon intensity, the carbon momentum, the green intensity, the ESG score, and the sector allocation for the optimized portfolios found in 3.(b) and 3.(c).

The following table summarizes the main characteristics of the optimized portfolios obtained in points 3.(b) and 3.(c).

Portfolio	TE Vol	CI	CM	GI	S	$w^\star(Sector1)$	$w^\star(Sector2)$
3.(b)	17.33%	219.2644	1.02%	5.63%	-0.7396	59.90%	40.10%
3.(c)	20.32%	149.4500	3.63%	3.86%	-0.8755	32.50%	67.50%

(e) Comment on these results.

Several remarks can be made.

First, the tracking error volatility is higher in portfolio 3.(c) than in portfolio 3.(b):

$$17.33\% < 20.32\%.$$

This is expected, since the addition of sector neutrality constraints makes the optimization problem more restrictive and reduces the set of admissible portfolios.

Second, the sector allocations differ substantially across the two cases. In 3.(b), the optimizer is free to move across sectors and shifts the portfolio toward $Sector_1$, allocating 59.90% to it. In 3.(c), by contrast, the sector weights are fixed at their benchmark levels by construction.

Third, the overall portfolio carbon intensity is lower in 3.(c) than in 3.(b), even though the same sector carbon-intensity targets are imposed in both cases. This is explained by the fact that, in 3.(c), the benchmark sector weights are preserved, and $Sector_2$ - which is much less carbon intensive than $Sector_1$ - keeps its relatively larger benchmark weight.

The results highlight the impact of sector-level decarbonization constraints and sector allocation constraints on the optimal portfolio.

Finally, both optimized portfolios exhibit a deterioration of the other ESG metrics (carbon momentum, green intensity, and ESG score). This reflects the fact that the optimization problem in point 3 focuses exclusively on sector-level decarbonization and sector allocation constraints. Since no additional constraints are imposed on the other ESG indicators, the optimizer is free to worsen them in order to minimize the tracking error.